

Anamika Saha

Software Engineer • Full-Stack Developer • Applied AI/ML • BI & Data Operations

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PROFILE

Computer Science graduate with experience in Full-Stack Development, Artificial Intelligence, Machine Learning and Business Intelligence & Data Operations. Proficient in Python, C++, Java, JavaScript, React.js, and Node.js, with hands-on experience in Oracle SQL, ETL processes, RESTful APIs and enterprise database management. Currently working as a Junior Software Engineer while pursuing research in AI and Machine Learning.

EDUCATION

Ahsanullah University of Science & Technology

B.Sc. in Computer Science & Engineering (CGPA: 2.871/4.00 | Last Two Years CGPA: 3.061/4.00)

2021 – 2025

Viqarunnisa Noon College

Higher Secondary Certificate (Science) (GPA: 5.00/5.00)

2018 – 2020

Siddheswari Girls' High School

Secondary School Certificate (Science) (GPA: 5.00/5.00)

2016 – 2018

- Government Scholarship Recipient

PROFESSIONAL EXPERIENCE

• BI & Data Platform Engineer

Jan 2026 – Present

Teletalk Bangladesh Ltd. (through Opus Technology Limited)

Gulshan, Dhaka

- Managed **Tableau Server monitoring** and **BI reporting** workflows to support telecom data-driven decision making.
- Worked with **Oracle databases** for tablespace monitoring, **SQL validation** and data loading using SQL Developer.
- Gained working knowledge of **Data Warehouse architecture** and **ETL pipeline** processes in telecom BI systems.
- Performed in telecom data analysis, including **CDR log monitoring** and basic CDR validation.
- Automated server maintenance tasks through **cron jobs** and **bash scripting**, optimizing pipeline flow and system monitoring.

PROJECTS

• Teletalk Subscriber Analytics Bot | Python, FastAPI, Llama 3.2, Oracle, ML

View Project

Built an AI-powered subscriber analytics chatbot for Teletalk Bangladesh featuring natural language querying and ML-based subscriber clustering.

• CDR14 Web Application | Python, Flask, Oracle SQL, PyQt5, HTML

View Project

Enhanced Teletalk's internal CDR investigation system with time-based filtering, Oracle SQL integration, a Flask web interface and CSV export.

• CDR Log Monitor Dashboard | Python, Flask, Paramiko, Multithreading, REST API

View Project

Built a real-time CDR pipeline monitoring dashboard for Teletalk tracking file registration, merge, and L1 loading stages across multiple CBS and MSC servers via SSH with live status updates.

• Clicon - eCommerce Full-Stack Application | React, Node.js, Express, MongoDB, JWT

View Project

Built a full-stack eCommerce platform with a React frontend and Express/MongoDB REST API, featuring JWT + Google OAuth authentication, product filtering, cart/wishlist/compare, and both guest and authenticated checkout with order tracking.

• Foodie | C#, ASP.NET MVC, MySQL, Visual Studio

View Project

Developed an online food ordering website featuring interactive menus, order management & responsive design.

• Interactive 3D Book Experience | Three.js, WebGL, Shaders

View Project

Developed a web-based immersive 3D book with realistic page-turning physics, custom shaders and 360° controls.

SKILLS

• Hard Skills:

Languages: HTML5, CSS3, JavaScript, TypeScript, Python, C, C++, MATLAB, Java(Basic), PHP, SQL

Frontend & Web: React.js, Next.js, Tailwind CSS, Bootstrap, Redux Toolkit, Material-UI

Backend & Cloud: Node.js, Express.js, BashScript, ASP.NET MVC, Flask, FastAPI, REST APIs, Uvicorn, Linux, AWS EC2

Databases: MySQL, MongoDB Atlas, Oracle SQL, Microsoft SQL Server, Firebase

Data & ML Pandas, NumPy, Matplotlib, Seaborn, TensorFlow, PyTorch, Scikit-learn NLP

Tools & Design: Tableau, Git, Docker, Netlify, Vercel, Figma, Canva

• Soft Skills:

Problem-Solving & Critical Thinking, Teamwork & Leadership, Technical Communication & Presentation, Time Management & Adaptability, Creativity & Project Management

RESEARCH & PUBLICATIONS

- **Cardiovascular Disease Prediction: A Machine Learning Approach** View Thesis
Thesis Defended; December 2025
Description: Researched predictive ML models using demographic, clinical and lifestyle data. Evaluated multiple algorithms including **LightGBM, CatBoost, XGBoost and ANN**, achieving **90%+ accuracy**.
- **Common Reasons for Chest X-ray Examinations in Hospitals: Understanding Causes, Diagnoses, and Raising Awareness in Dhaka, Bangladesh** View Journal
Journal Published; European Journal of Medical and Health Sciences; September 2025
Description: Analyzed the clinical reasons for chest X-ray examinations across hospitals in Dhaka, evaluating diagnostic outcomes and public awareness to identify trends and support improved healthcare practices.
- **Artificial Intelligence in Chest Radiography Comparative Study with Hospital Radiologists' Reports in Dhaka, Bangladesh** View Journal
Journal Published; European Journal of Artificial Intelligence and Machine Learning; November 2025
Description: Compared a DenseNet-121 AI model with hospital radiologists' chest X-ray reports to evaluate diagnostic agreement, identify reporting differences, and assess AI's potential as a clinical decision-support tool.
- **A Deployment and Resource Aware Edge AI Framework for Real-Time Waste Classification with Biometric Authentication and Secure Logging** View Work
Manuscript under review; PLOS ONE; submitted April 2026
Description: Engineered an automated waste management system integrating **Edge-AI** for real-time classification, **Biometric Authentication** and a **Blockchain-based incentive mechanism** to ensure data integrity and transparency.
- **ZETA: A Zero-Knowledge and Threshold-Cryptographic Architecture for Scalable and MEV-Resilient Sharded Blockchains** Manuscript
Manuscript under review; Engineering Reports (Wiley); submitted July 2026
Description: Proposed a scalable blockchain architecture integrating Zero-Knowledge Proofs, Threshold Cryptography and sharding techniques to improve security, transaction throughput and resistance against MEV attacks.

TRAINING / CERTIFICATIONS

- **Automation Essentials: Industrial Process Control System Design (AUSTRC)** 30 – 05 – 2024
- **Hands-on Workshop on FPGA Acceleration for Machine Learning Applications** — View Certificate 30 – 01 – 2025
- **Workshop on GitHub, CNCF and AWS – AUST Programming and Informatics Club** — View Certificate 26 – 08 – 2025
- **Space Webinar on “Charting Saturn’s Secrets: The Cassini Mission & Beyond”** — View Certificate 23 – 01 – 2025

EXTRA-CURRICULAR ACTIVITIES

- **WebMaster** July 2025 – Dec 2025
AIAA AUST Student Branch — View Certificate
Maintained the branch's official website and technical infrastructure.
Hosted and anchored various online and offline programs and meetings on behalf of the branch.
- **Deputy Executive of GFX Team** May 2024 – Aug 2025
AUST Robotics Club — View Certificate
Led the design team to create event posters and digital content; **Co-led** Guests and Judges Handling for **ARC 2.0** (AUST Rover Challenge).
Served as an organizer, host and anchor for the National Robotics Event's Prize-Giving Ceremony.
- **Graphic Designer** 2024 – 2025
Satellite Communication Laboratory, AUST
Designed posters, infographics and visuals for internal and external lab communication.
- **Campus Ambassador** Oct 2024 – Nov 2024
Robo Tech Valley — View Certificate
Worked as a Campus Ambassador for the **4th International Robo Tech Olympiad 2024**.
Promoted robotics and IT programs among university students through outreach and awareness campaigns.
- **Participant, Volunteer & Organizer** Spring 2022 – Spring 2024
AUST CSE Carnival
Volunteered in CSE Carnival 1.0 to 3.0; Participant in CSE Carnival 2.0 to 4.0.
Organizing Committee Member for CSE Carnival 5.0.
- **Sports & Event Organizer** Spring 2021 – Fall 2024
AUST CSE Department
Represented the department in Football and Cricket tournaments across all semesters, serving as Football Team Captain (Fall 2021).
Organized major departmental events including CSE Cricket Tournament, CSE Picnic and CSE Revelry (CSE Fest) - Spring'24.
Awarded Man of the Match - Spring'22 (Cricket) & Fall'23 (Football)

REFERENCES

- **Ms. Syeda Shabnam Hasan** – Assistant Professor, Dept. of CSE, AUST – shabnam.cse@aust.edu
- **Mr. Saha Reno** – Assistant Professor, Dept. of CSE, AUST – reno.cse@aust.edu